

HandAttNet: Attention 3D Hand Mesh Estimation Network

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ABSTRACT

Hand pose estimation and reconstruction become increasingly compelling in the metaverse era. But in reality hands are often heavily occluded, which makes the estimation of occluded 3D hand meshes challenging. Previous work tends to ignore the information of the occluded regions, we believe that the occluded regions hand information can be highly utilized. Therefore, in this study, we propose hand mesh estimation network, HandAttNet. We design the cross-attention mechanism module and the DUO-FIT module to inject hand information into the occluded region. Finally, we use the self-attention regression module for 3D hand mesh estimation. Our HandAttNet achieves SOTA performance.

1 INTRODUCTION

With the rapid development of deep learning in computer vision significant progress has been made in estimating 3D hand lattices from a single RGB image [2, 4, 6], however, the method for estimating 3D hand lattices with occlusion is still an open challenge. And a more advanced approach for now is to use a spatial attention mechanism to obscure the robustness problem. This method is currently used in the problem of estimating 2D human pose [3, 10] for an occluded body. Although the approach based on spatial attention mechanism has achieved good results in solving the occlusion problem, it still has some limitations. To overcome these limitations, we propose a new occlusion robust 3D hand mesh estimation framework HandAttNet. As shown in Fig. 1.

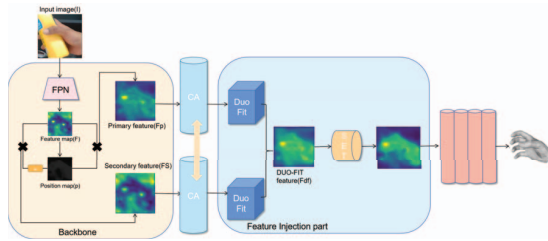


Figure 1: The overall architecture of HandAttNet

On the one hand our cross-attention [1] mechanism is able to exploit not only the intrinsic relationship between the two features extracted by the backbone feature, but also the correlation between the image region and another feature in order to complement and enhance each other's generated features. On the other hand in order to enhance our injected features, we use transformer to fully consider the feature connections between contexts. Unlike the previous FIT feature injection methods [8], we also add a processing of the primary features, and the primary features obtained by using Duo-Fit processing take full advantage of the feature linkage between the

primary and secondary features, not only injecting the information of the primary features into the region of the secondary features. At the same time, the information of secondary features is injected into the region of primary features. Then the two feature mappings are fused to obtain a new single feature input to the SET network later. Finally, we apply the self-attentive mechanism module to the final regression module of the network, which reduces the dependence on external information and allows a better performance on the internal relevance of the returned features.

We demonstrate the effectiveness of our handAttNet through extensive experiments on recently released hand-object interaction datasets (e.g., HO-3D [5]). The experimental results show that our HandAttNet obtains better 3D hand mesh estimation accuracy compared to previous state-of-the-art 3D hand mesh estimation methods. To summarize, we make the following contributions:

- We propose a new framework, HandAttNet, for occlusion robust 3D hand mesh estimation from a single RGB image. The proposed handAttNet utilizes a cross-attention mechanism as well as a dual feature injection mechanism to make the feature mapping robust to occlusions by properly injecting hand information into the occluded region.
- For the new network structure, we propose the cross-attention mechanism to process the primary and secondary features. The DUO-FIT module use feature injection and then send into the SET for feature refinement. Finally the network training is completed using a regression module with a self-attention mechanism.
- We demonstrate that our framework significantly outperforms state-of-the-art 3D hand mesh estimation methods on hand-object interaction datasets containing severe hand occlusion

2 METHOD

In Figure, we provide an overall pipeline of our HandAttNet for 3D hand mesh estimation. Our HandAttNet consists of backbone, cross-att, duo-FIT, SET and att-regressor.

2.1 Backbone

The backbone extracts feature F and Position map P from a hand images $I \in R^{512 \times 512 \times 3}$. We first feed the hand image I to FPN, which based in ResNet50 and re-size the output feature map $F \in R^{512 \times 512 \times 3}$. Then, we use convolutional layers and then use the sigmoid function to estimate the Position map P from the feature map F . According the score of the Position map, we separate the feature map F into primary feature F_p and secondary feature F_s .

2.2 Cross attention(CA)

The cross-attention mechanism module further processes the backbone features to extract the primary and secondary features generated by post-processing, and the features are fused by attention several times to complement each other, making full use of each tokens to query with other tokens to finally form better primary and secondary features. For the inputs of CA module are the $F_p \in R^{3 \times 256 \times 256}$ and $F_s \in R^{3 \times 256 \times 256}$ and the shape of outputs are same with inputs. By using F_p 's and F_s 's value and query and key, to fully exploit feature information.

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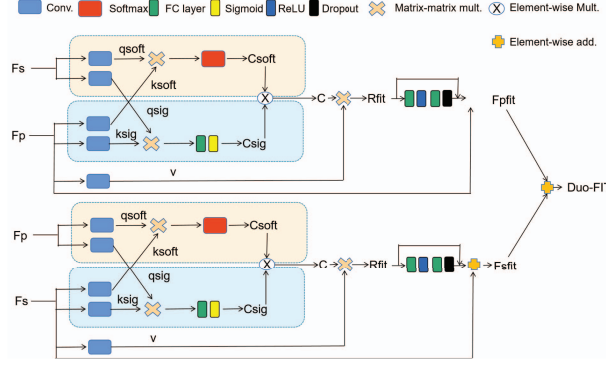


Figure 2: Duo-FIT

2.3 Duo feature injecting transformer(Duo FIT) and SET

The schematic diagram of Duo-FIT is shown in Fig2. Duo-FIT is a transformer-based [7–9] module that employs two features, F_p and F_s , and injects information of both features into each other by considering their correlation. We use two sub-modules in Duo-FIT, the softmax-based attention module and the sigmoid-based attention module. SET improves the features by referencing the distance information of the features and by self-focusing.

The following equations will show the softmax attention module and the sigmoid attention module: where pool denotes average pooling, C means correlation map, q means query, k means key, d_k means the feature dimension of the key

$$C_{soft} = \text{softmax} \left(\frac{q_{soft} k_{soft}^T}{\sqrt{d_{k_{soft}}}} \right) \quad (1)$$

$$C_{sig} = \text{sigmoid} \left(\text{pool} \frac{q_{sig} k_{sig}^T}{\sqrt{d_{k_{sig}}}} \right) \quad (2)$$

Input of DUO-FIT are the treated F_p and F_s from CA. And after the module Duo-Fit, the feature become fusion. The outputs are the features $F_{DUO-FIT} \in R^{3 \times 256 \times 256}$.

$$F_{DUO-FIT} = F_{P-FIT} \oplus F_{S-FIT} \quad (3)$$

2.4 Attention regressor

The regressor generates MANO pose and shape parameters, and forwards the MANO parameters to the MANO layer to obtain the final 3D hand mesh. we add the self-attention module in regressor in order to enhance the Generated pose and shape parameters. to get the final 3D hand joints coordinates and obtained the final 3D hand mesh $V \in R^{778 \times 3}$

3 EXPERIMENTS

We used four 3090 GPUs for training and all implementations were done with PyTorch. We trained on HO-3D [5], each model was trained with a learning rate of 10-4 every 10 epoch. all other details will be show in our code.

Table 1: Output error results

Methods	Joint	Mesh	F@5	F@15
Pose2Mesh [2]	12.5	12.7	44.1	90.9
HandOccNet [8]	9.1	8.8	56.4	96.3
Our	8.5	8.2	57.8	97.0

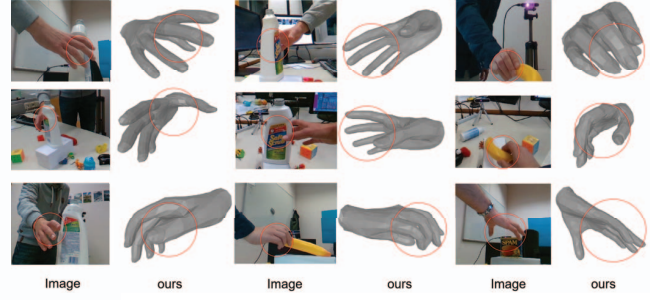


Figure 3: Output mesh

4 CONCLUSION

We propose HandAttNet, a framework for 3D hand mesh estimation in the presence of occlusion. Our HandAttNet uses the attention mechanism and cross-feature injection mechanism to make full use of the primary and secondary features generated by feature extraction to achieve 3D hand mesh estimation in the presence of occlusion. We have completed the construction of the cross-attention module and the DUO-FIT module, and verified the effect of adding these modules for 3D hand mesh estimation.

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